

APEX TITAN SYSTEM ARCHITECTURE

Technical Documentation & Integration Blueprint

1. Titan Node (Android Application)

- MainActivity: The Secure Hub. Manages Biometric Shield (Fingerprint/Face) and the Hybrid Bridge.
- TitanBridge: A high-velocity JavaScript Interface connecting Native Hardware to Web UI.
- Edge AI Scanner: ML Kit powered engine for Barcode Scanning, AI Triage, and Shelf Auditing.
- LocationService: Persistent foreground synchronization for real-time node tracking.
- Supabase Integration: Encrypted storage for offline drops and REST API configuration sync.

2. Apex Command Hub (Next.js / Web)

- Intelligence Hub: Real-time PostgreSQL signal monitoring and tactical analytics visualization.
- Mission Control: Centralized logistics management for dispatching orders and tracking nodes.
- Framework: Built with Next.js 15, optimized for deployment on Render, using Framer Motion for UI.
- Deployment: Continuous integration via GitHub 'tech' remote.

3. System Interconnectivity & Data Flow

- Native-to-Web: Titan Node injects 'TitanNode' object into WebView for immediate hardware access.
- Web-to-Native: The Dashboard triggers evaluateJavascript() to sync scanning results back to the UI.
- Cloud Synchronization: Both platforms share a unified Supabase backend for real-time mission data.
- Security Handover: Dynamic QR generation enables encrypted physical-to-digital rider validation.